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Integration of digital twin and IoT for virtual greenhouse management

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ABSTRACT

The convergence of Digital Twin (DT) and Internet of Things (IoT) technologies represents a key enabler of Agriculture 5.0, providing real-time synchronization between physical and virtual environments. When coupled with Virtual Reality (VR), these technologies enable immersive and interactive systems that enhance intelligent agriculture monitoring, control, and decision-making. This study proposes an advanced and fully integrated VR–DT–IoT framework for virtual greenhouse management, designed not only to simulate and visualize but also to synchronize and control the physical system in real time. The proposed framework achieves seamless cyber-physical fusion by establishing bidirectional communication among IoT sensors, DT models, and VR interfaces. This enables users to trigger actuator commands and observe instant physical feedback directly within the immersive environment. A real-world experimental greenhouse was developed as a testbed to validate the system's architecture and functionality. IoT sensors continuously monitored environmental parameters with data streams dynamically synchronized to the DT model reproducing greenhouse dynamics. The VR interface provided context-aware interaction and spatial visualization, allowing users to explore live sensor data, analyze trends, and simulate adaptive control strategies in a realistic virtual environment. The experimental validation, conducted with 23 participants across seven training sessions, revealed a marked reduction in task completion time and error rates, along with high usability scores (SUS = 3.84/5), confirming the system's effectiveness in improving operational performance. The findings demonstrate that the proposed system significantly advances digital agriculture by merging modeling, real-time feedback, and immersive control into a unified human-in-the-loop platform. This research highlights the technological novelty and practical value of integrating DT, IoT, and VR into a single coherent framework, bridging physical and virtual domains.

Introduction

Climate volatility, resource scarcity, and management complexity in controlled-environment agriculture revealed a huge gap between the abundant sensor data and the understandable, timely decisions for sustainable crop management [1]. The issue is aggravated everywhere especially in Africa, where crop productivity is still highly dependent on environmental variability, resource-wasteful application, and limited access to high-end decision-support technology. As a result of such challenges, it is

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necessary to have technological innovation and synchronization with broader development agendas that foster capacity, resilience, and digitalization in agriculture [2].

This research presents a Virtual Reality–based Digital Twin–IoT (VR–DT–IoT) architecture as a means to bridge the gap. It provides real-time synchronicity between the physical greenhouse and its digital twin, thereby enabling the real-time monitoring, prediction, control, and closed-loop automation [3]. Besides its technological novelty, the integration is a practical solution to one of the most African problems—how to manage agricultural resources effectively and sustainably in an uncertain climate. Supported by an interactive human–machine interface, the system supports experiential training, situation awareness, and decision rehearsing in risk-free virtual environments, positively affecting operational effectiveness and human capacity development.

Most recent work addresses modeling, sensing, and visualization as distinct modules and allocates reproducibility, temporal consistency, and user interaction. Recent approaches are more likely to direct attention to data acquisition or simulation fidelity but less likely to achieve a complete end-to-end system that (i) streams mixed sensor data, (ii) maintains an updated virtual greenhouse in real time, and (iii) estimates the impact of operator decisions within a group feedback loop. Besides, conventional training processes are costly, dangerous, and difficult to replicate. In contrast, simulated electronic environments allow operators to try out critical responses for free without damaging crops or machinery.

Deep sensing and communications infrastructure is needed to make such an integration possible, serving as the basis for real-time synchronizing and intelligent decision-making. For this purpose, Internet of Things (IoT) facilitates constant acquisition from the physical world and safe transportation to the digital world, where multi-sensor information like temperature, humidity, water content in soil, light intensity, pH, and carbon dioxide (CO_2) content is conditioned for adaptive control and evaluation [4]. The sensor networks and embedded systems thus allows one to engage in precision agriculture and optimize that use of inputs [5]—two goals that align with the stated Agenda 2063 Goals 5 (Modern Agriculture for Increased Productivity) and SDG 2 (Zero Hunger) and 6 (Clean Water and Sanitation) through data-driven irrigation, energy-efficient practices, and effective crop management using evidence and insights [6]. However, undeniably, the second half of the task is working with the data; there is a well-founded need for advanced capabilities to process, plot, and report the data into useful information [7].

Some mathematical frameworks have appeared to characterize dynamic systems that allow for decision-making under conditions of uncertainty. In addition to optimal control theory and Reproducing Kernel Hilbert Spaces (RKHS) [8], there are mathematical frameworks to model nonlinear dynamic systems [9] and ordinary differential equations (ODEs) [10], as well as, recurrent neural networks (RNNs) [11], as well as hybrid models that use both fuzzy logic and deep learning [12], some of which present possibilities for use as Digital Twin (DT) modeling because of the enhanced certainty of simulating and predicting system dynamic as closely to real-time scenarios as possible. These frameworks afford more accurate and predictable analyses of DT systems. However, their revolutionary nature is leveraged upon application of virtual realities, and/or IoT, thus, allowing for entirely new paradigms of systems control and systems engagement [13,14]. A DT is initially a framework for cyber–physical integration that provides a real-time, high-fidelity virtual replica of a physical object—a system, component, or process—made available through sensor networks [15], actuation mechanisms, and bidirectional communication frameworks [16].

It usually consists of three interconnected elements: (1) the Physical Space, which is the physical object that has physical sensors and actuators; (2) the Virtual Space, which is comprised of a computational model that emulates the form, function, and status of the physical object; and (3) the Data Link, which maintains a consistent sync between physical space and virtual space by employing mechanisms for data acquisition and control transfer [17]. The physical-to-virtual pipeline generally builds on state acquisition from the physical world in real-time using IoT-sensing-capable modules, which are passed via lightweight protocols to a virtual model during monitoring, state estimation, or simulation. Alternatively, the virtual-to-physical pipeline transfers control commands or optimized setpoints back to the physical object, where actuators will make the appropriate adjustments.

Most DTs deploy edge–cloud infrastructures, semantic data models, and multi-physics simulations in real-time, and they are scaled to respond. A DT's performance depends on the model fidelity and temporal consistency (sync latency between the physical and virtual domains). While DTs strongly emphasize manufacturing industries and, to a broader extent, the industrial sector, DT has not yet been utilized with IoT and Virtual Reality (VR) in agricultural sector. Most recent studies focus on IoT-based environmental monitoring or computer simulation, but with no real-time bidirectional interaction or immersive visualization of the physical process. With increasingly sophisticated and data-intensive agricultural production systems, a greater need exists for data collection and analysis frameworks with interactive, human-centered environments for decision-making, training, and system control. As a part of this larger vision, the presented framework not only advances the state-of-the-art in innovative greenhouse technologies but also provides a scalable digital agriculture model aligned with African and global sustainability agendas, by providing an innovative and highly integrated VR–DT–IoT framework.

The platform establishes persistent, real-time synchronization between the physical greenhouse and its 3D digital twin, whereas virtual reality interface enables immersive visualization and bidirectional control of actuators and processes. The integration introduces a new degree of interactivity and gives-and-take that transforms passive data representation to active experience-based management and elevates the user's comprehension and interaction level. The approach bridges the gap between virtual and physical, moving toward a better improvement that is key to Agriculture 5.0. This article has four primary contributions. (1) It suggests an integrative Digital Twin–IoT framework for virtual greenhouses that unifies sensing, data streaming, modeling, and actuation into a closed-loop cyber-physical system (see Supplementary Figure S1). (2) It sketches an implementation-oriented implementation with an embedded IoT sensing layer, a real-time DT, and a VR-based immersive user interface for visualization and interaction. (3) It describes a greenhouse case study contrasting operational understanding and decision performance with the immersive interface to monitoring. (4) It shows design trade-offs and concludes scalable and sustainable deployments in managed agricultural environments. We aim to demonstrate that a rigorously synchronized DT–IoT pipeline supplemented with interactive VR immersion can considerably improve

operator comprehension, knowledge recall, and decision-making effectiveness while enabling resource-efficient control in greenhouse uses.

Related work

The combination of IoT sensing and DTs allows for the monitoring of greenhouse conditions in real-time to establish the data basis, create low-resource crop climates, and timely interventions [18]. Howard et al. developed a Greenhouse Industry 4.0 DT model that combines IoT with AI to optimize energy consumption and productivity without compromising on quality or sustainability [19]. A wider view places expectations on DTs to boost intelligent farming through remote control, autonomous driving, and simulation of scenarios, amplifying climate response and crop care in controlled environments [20]. Platforms for DT–IoT based on cloud computing provide the scalability necessary to bridge small- and large-scale farms [19]. VR adds natural and interactive interfaces to DTs. Bernetti et al. (VAIFARM) combined social VR and generative AI for learning and planning farm management [21]. In the manufacturing sector, VR and DT combinations have showcased visualization of processes, machine operation, and co-simulation of human–robot work cells (Unity3D) with the potential for excellent synchrony and ergonomic analysis [22]. Horticulture has also seen user studies of virtual greenhouse twins highlight practical issues, including simulation sickness and previous gaming experience, that set the conditions for evaluation and the purposes of designing systems [23]. These conclusions lend credence to the use of VR for interpretation, training, and discovery purposes in agri-targeted DTs. Similarly, advanced analytics adds further strength to DT pipelines. Machine-learning-based detection of crop deficiencies and diseases can be integrated with DT monitoring for timely action; related mathematical modeling informs complex bio-process simulation [24]. Recent work on the Fast Plant Digital Twin leverages Neural Radiance Fields (NeRF) to cost-effectively scale 3D plant morphology modeling in IoT-enabled greenhouses [25]. Low-cost DT–IoT frameworks such as AgriLoRa combine embedded vision with mobile models (MobileNet, U-Net) for disease/weed detection at high precision [26].

Beyond technology, adoption studies grounded in the Technology Acceptance Model identify self-efficacy, social norms, and system quality as key drivers of perceived usefulness and ease of use in agriculture [27].

Despite these advances, convergence across VR, DT, and IoT technologies remains limited in the agricultural domain. Most existing

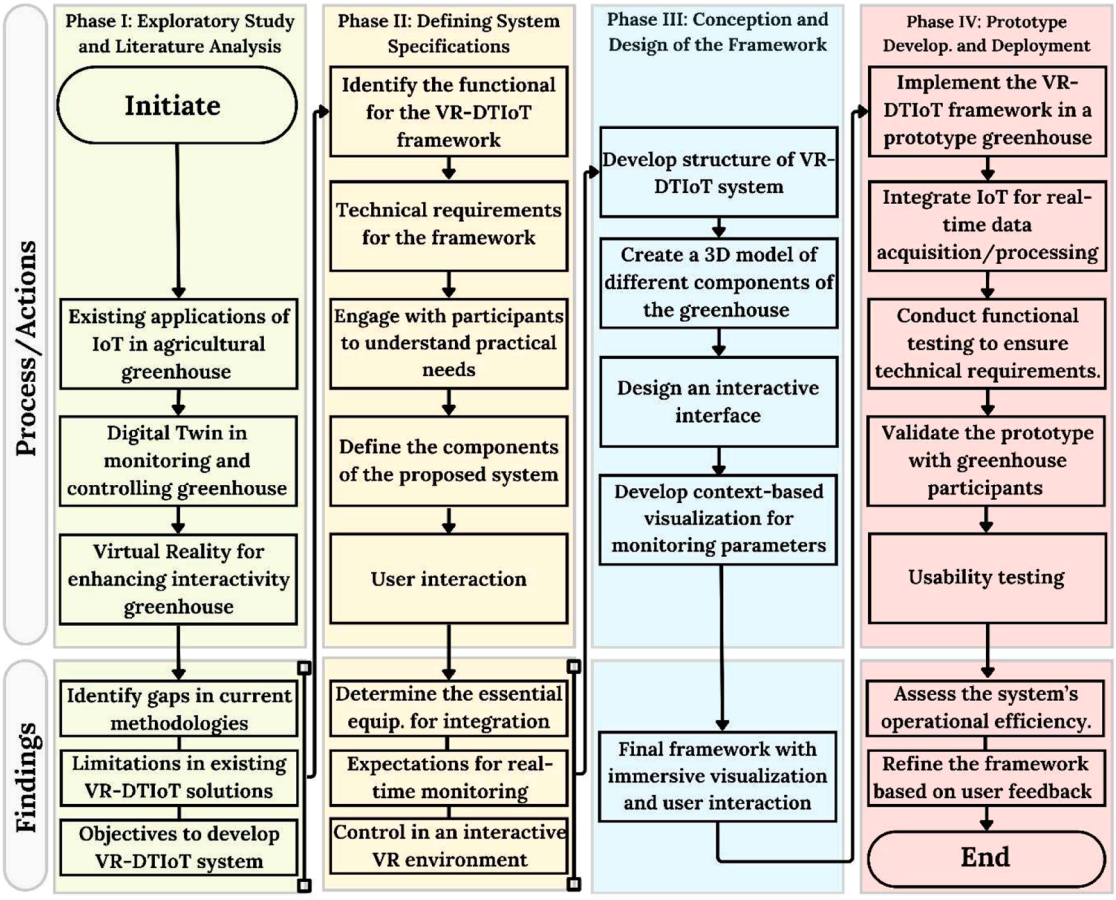


Fig. 1. Research methodology framework: phases of development and validation.

studies address either sensing, simulation, or visualization in isolation, with few achieving a fully integrated, closed-loop pipeline capable of (i) streaming multi-modal data in real time, (ii) dynamically synchronizing a virtual greenhouse with its physical counterpart, and (iii) assessing how operator decisions influence future system states within a unified framework.

Our literature mapping (2020–2024) confirms that while agriculture is covered (see Supplementary Figure S2), genuine smart agriculture — as characterized by real-time interactivity, human-in-the-loop integration, and cyber-physical feedback — is understudied. This convergence of technologies is only sporadically investigated, occasionally outside agriculture or in scaled-down prototypes lacking immersive interaction or bidirectional control capability [23]. Chronic bottlenecks identified are data interoperability, representation heterogeneity, and absence of intuitive and ergonomic interfaces co-designed with end users for real-world uptake [28]. Such discrepancies clearly demand an integrated VR–DT–IoT approach bridging the physical and virtual worlds by real-time synchronization, predictive modeling, and immersive interaction. The strategy in this paper rather differently layer it is a direct remedy of these kinds of limitations that somehow also unifies environmental sensing, digital twin simulation, and interactive visualization into one unified form.

In comparison to the earlier paradigms that were merely limited to passive observation or static modeling, our system presents a new cyber-physical integration paradigm where users can perceive, control, and get accustomed to greenhouse operations in an interactive virtual world.

Materials and methods

The methodology employed in this research is largely based on Design Science Research (DSR) principles and follows a systematic and iterative four-stage framework: (1) literature review and initial system architectural design; (2) determination of system specifications from functional and technical requirements; (3) the invention and detailed design of the VR-DT-IoT framework proposal; and (4) prototype creation, confirmation, and performance measurement. Every stage was carefully crafted to conserve the methodological rigor, technical feasibility, and applicability of the agricultural greenhouse environments of the real world. Fig. 1 presents the research methodology framework, which depicts the major stages involved in the creation and confirmation of the proposed system.

Architecture design and functional specification

The approach in this study addresses these challenges by joining environmental sensing, digital twin simulation, and interactive visualization into one system. The work began with a detailed requirements analysis to design a VR–DT model suited for greenhouse management.

An IoT sensor network collected agroclimatic data. A central gateway aggregated the data streams and performed initial processing. The gateway was selected for its low cost and computing efficiency, and it matched the needs of an IoT-based immersive setup. Node-RED acted as middleware between sensors, processors, and VR–DT interfaces. It handled data processing and coordination among component nodes through filtering, normalization, and formatting to ensure data consistency [29].

Communication between the physical and virtual systems worked both ways through two separate links. The Physical-to-Virtual link transmitted sensor data to the VR–DT environment for visualization and analysis. In contrast, Virtual to Physical (VPc) dispatched control signals and generated commands to physical actuators. Data in real time was communicated using MQTT over a HiveMQ broker; the publish-subscribe model allowed messages to be decoupled and was optimized for low-latency messaging under high connection counts and load conditions. Node-RED orchestration and MQTT messaging provided a secure, fault-tolerant

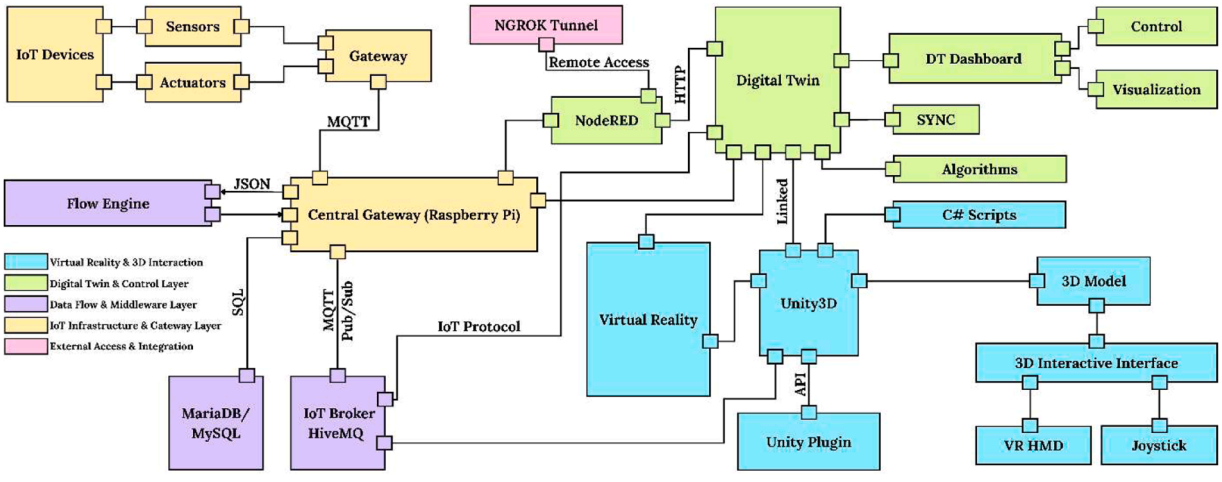


Fig. 2. Architecture of the integrated VR-DT-IoT framework for smart greenhouse management.

connection architecture to connect the greenhouse with a VR-based replica system [30]. The stakeholder input from agronomists and greenhouse operators clarified real-time monitoring requirements, alert-like thresholds, decision support, and simple navigation in the VR interfaces that aligned the system with operational support needs. These features and components represent the framework of the VR-DT that will be built upon to include the broader context of integrated greenhouse management in future phases of the research. Fig. 2 outlines the proposed VR-DT-IoT architecture, with color-coded layers separating infrastructure, middleware, simulation, and user interface.

VR-DT-IoT modeling

In this research phase, we developed and constructed a VR-DT-IoT model utilizing Unity3D software. This was a key step in establishing an interactive, dynamic, and immersive environment that merged the data from real-world sensors into an immersive virtual world to guarantee the efficient management of a greenhouse (Fig. 3a & b). Several significant decisions were made during the development, each of which was crucial for the technical as well as the functional aspects, to make the solution strong and easy to use. A very detailed 3D model of a greenhouse was made in Unity3D, with the structural and functional aspects all taken into account, in order to simulate the physical world as realistically as possible. The model has main pieces: walls, benches, irrigation, ventilation systems, lights, and rows of crops. Besides, great extra effort was made to render the model visually realistic and rich in terms of semantics. Moreover, each virtual component was linked to the respective sensory data point creating a semantic link. Semantic integration allowed virtual objects to reflect changes in real-time data dynamically, thus making the whole system more engaging and reliable.

Greenhouse simulation was tuned with modern rendering capabilities and level-of-detail approaches to ensure scalability and file size within reasonable limits. System, sensor, actuator, and digital twin setup initialization is specified in Algorithm 1. This pseudocode is the initial setup required to prepare the system for real-time usage in a suitable manner. Since most common VR configurations are used, this allows it to run well even on systems with low processing capacity. Integrating real-time sensor data with the virtual model

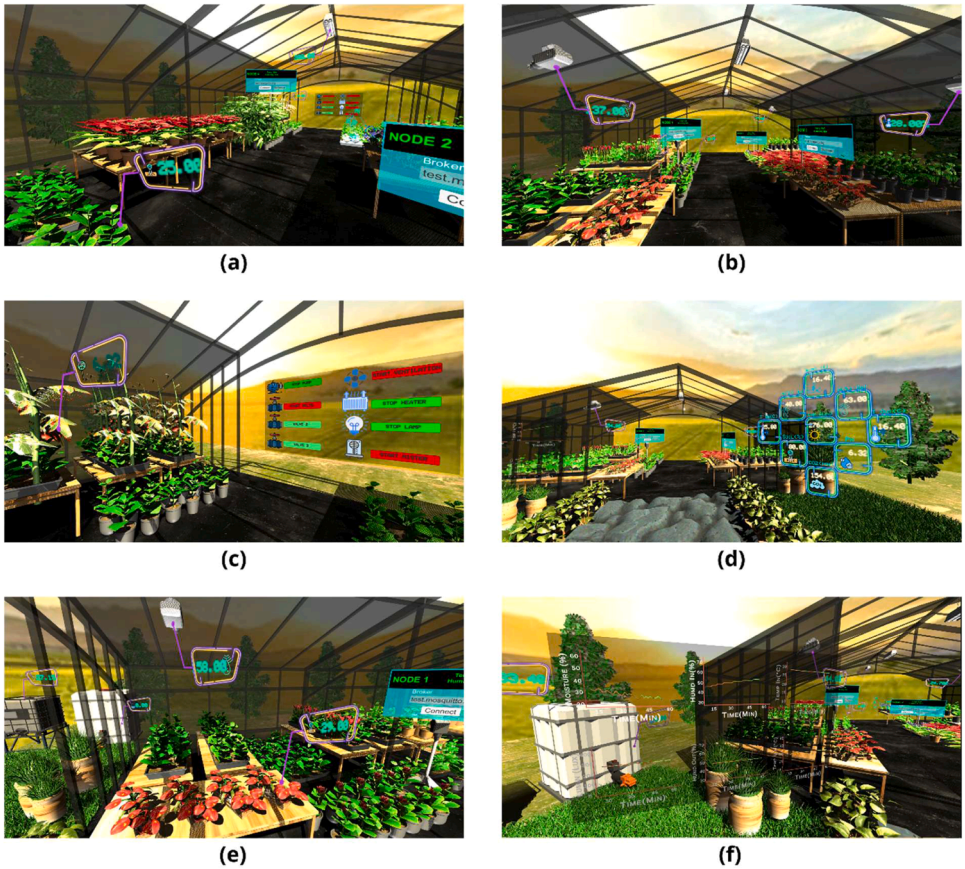


Fig. 3. Virtual reality user interfaces for VR-DT-IoT greenhouse operation: (a–b) Navigation and interaction, (c) Actuator control with live feedback, (d–e) Visualization of user interactions with greenhouse components, and (f) Interactive chart displaying environmental sensor data over the last 60 min.

was at the core of the system. Real-time processing data for seamless synchrony of sensor inputs and the VR model is given in Algorithm 2. The process explains how the sensor data streams obtained from the central gateway are processed and dynamically mapped to their corresponding virtual components.

Unity3D scripting provided natively integrated MQTT support for real-time DT updates. Scripts subscribed to HiveMQ topics, read sensor streams, and updated the twin in real-time, allowing real-time representation of environmental change in the VR scene [31]. We employed a uniform set of navigation and control paradigms to provide intuitive, cross-device interaction. The navigation, inspection, and actuation workflows are outlined in Algorithm 3. The users move around the greenhouse, look at detailed readings, and engage with virtual components using chart visualizations (historical trends), actuator commands, irrigation management, and plant-growth monitoring dashboards.

User selects a visualization mode, enters a greenhouse compartment by gestures/touch/joystick, and points at equipment or sensors to open a contextual control panel with status and controls. Actions on selection gestures are to actuate, set environmental setpoints, or fetch history; commands are transmitted via MQTT to the backend for real-time synchronization. Visual feedbacks complete the loop. For rapid diagnosis, click-to-view data points expose current data, and time-series views. Motion controller-based gesture navigation allows for fluid motion, zoom, and pan, with increased immersion and responsivity of the VR-DT interface [32].

On the other hand, context-sensitive menus appear dynamically as the user is dealing with some elements of the greenhouse. These provide options for displaying detailed information of the greenhouse, adjusting environmental breakpoints, or even triggering alarms efficiently by providing contextually relevant options [33]. Within this hierarchical data visualization system, the data is grouped and presented simply and comprehensively to prevent information overload and ensure easy access.

The visualization layer of the system has been designed to visualize greenhouse data comprehensively and interactively, allowing users to see and understand important data in real time. High-level summary statistics – such as average temperature, cumulative water use, and overall greenhouse efficiency – are presented to users accessing the interactive dashboard to advance recognition of key operating statistics. Creating a low-tech, interactive user interface also enabled the users to survey and manage greenhouse activity satisfactorily through this interface. The linking of real-time visualization, interactive user interface, and high-performance data visualization yielded an enacted VR-enhanced DT model. These elements were backed by a solid validation of the existing control capabilities of the greenhouse, creating a sound basis for the implementation and validation phases and verifying that a competent technical infrastructure was achieved to test user needs [34].

Algorithm 1: System Initialization and Readiness Setup for the VR–DT–IoT Greenhouse Platform

Data: `Sensors`, `Actuators`, `MQTTBroker`, `Database`, `Thresholds`
Result: `System readiness for real-time operation`
`/* (1) Sensor and Actuator Initialization */`
`Sensors` ← **Initialize**({Temperature, CO_2, pH, …}) at strategic locations
`Actuators` ← **Deploy**({Pump, Irrigation, Ventilation, Mister, Lighting Systems, …})
Ensure functional connectivity with all control units
`/* (2) System Configuration */`
`RaspberryPi` ← **Configure** as the central gateway for data acquisition and preprocessing
`MQTTBroker` ← **Setup**(`HiveMQ`, QoS=1, Topics = [“VR-DTIoT_GH/Sensors/#”, “VR-DTIoT_GH/Actuators/#”])
`Database` ← **Setup**(Structure = {SensorReadings, ActuatorCommands, Alerts})
`Thresholds` ← **SetOptimalRanges**(Sensors, Actuators)
`EscalationProtocol` ← **DefineSteps**(Notify Operator → Log Event → Trigger Actuator)
`/* (3) Initialization Completion */`
`Sensors` → Begin transmitting data to `RaspberryPi` via `MQTTBroker`
`Actuators` → Set to standby mode
`DigitalTwin` ← **SyncInitialValues**(`Sensors`)
`VR.Environment` ← **LaunchAndRender**(GreenhouseModel, Mode=RealTimeOverlays)
return *`System readiness for real-time operation`*

System implementation and integration

The implementation focused on integrating hardware–software components into a real-time VR-DT pipeline. The computing platform (Intel® Xeon® W-2223 3.6 GHz, 16 GB RAM, NVIDIA Quadro P1000) was sufficient for the proposed model workload. The prototype was built in Unity 2020.1.8f1; greenhouse structures (walls, benches, ventilation, LEDs, valves, pumps) were modeled in Blender 3.6.17 and imported via FBX to preserve geometry, materials, and metadata. For user studies, the scene ran on an HMD; custom Unity scripts mapped controller inputs to in-scene actions. In the pilot greenhouse, calibrated sensors captured key variables. A Raspberry Pi 4 was the gateway, forwarding streams to a Node-RED server. Flow definitions handled acquisition, noise filtering, value standardization, and publication to a HiveMQ broker. MQTT provided scalable distribution.

Unity scripts subscribed to topics and updated the digital twin on each sensor change, leveraging the engine’s rendering to maintain an up-to-date scene. Command acknowledgments and on-screen cues closed the feedback loop for interactive control. Calibrated sensing, lightweight gateway, Node-RED preprocessing with routing in JSON, MQTT brokering, and Unity-based synchronization enabled reliable, real-time coupling between the physical greenhouse and its virtual twin. The data paths and integration strategies linking the IoT platform with the VR-DT environment are illustrated in the supplementary material (see Supplementary Figure S3) [35].

Algorithm 2: Real-Time Data Processing and Synchronization between IoT Sensors, Digital Twin, and VR Environment

```
Data: SensorData.Live, HistoricalData
Result: Updated Digital Twin, Alerts, Trends
/* (1) Continuous Data Loop                                          */
while SystemActive do
    /* (1.1) Data Acquisition and Preprocessing                      */
    SensorData.Live ← Fetch(MQTT.Subscription, Topic = "VR-DTIoT_GH/Sensors/#")
    SensorData.Raw ← Collect(Sensors)
    SensorData.Clean ← Preprocess(SensorData.Raw, Filter = "", Normalize = True)
    /* (1.2) Routing and Storage                                     */
    NodeRED.Input ← SensorData.Clean
    NodeRED.Route(Input = SensorData.Clean, Output = {DTModel, VRInterface})
    HistoricalData ← Retrieve(DB = "Local_Database", Range = "Last30Days")
    /* (1.3) Dynamic Digital Twin Update                             */
    foreach DataPoint ∈ SensorData.Live do
        DTModel.ComponentValue ← Map(DataPoint.Location, DataPoint.Value)
        if DataPoint.Value ∉ Thresholds[Component] then
            Highlight(Component in DTModel, Style = ColorChange)
            AlertPanel ← Display(Details = {Component, CurrentValue, RecommendedAction})
        end
    end
    /* (1.4) Anomaly Detection                                       */
    if SensorData.Delay > Threshold then
        Mark(DataPoint, Label = "Delayed")
        TriggerFallbackMeasures(Method = "UseLastKnownValue")
    end
    if SensorData.Missing == TRUE then
        NotifyOperator(Issue = ''Sensor Communication Failure'')
        LogError(For = MaintenanceReview)
    end
    /* (1.5) Trend Analysis                                          */
    Trends ← Analyze(HistoricalData, Tool = "Node-RED")
end
return Updated Digital Twin, Alerts, and Trends
```

Results

Case study

The architecture comprises tightly coupled entities spanning physical and virtual domains (see Supplementary Figure S4). The environmental sensor readings from the greenhouse paired with the actuator states are in a cyber-physical loop with their real-time virtual twin for interaction and visualization. The IoT gateway, the sensing/actuation layer integrator, also provides inputs to the DT Server, the computational hub. The DT Server can support algorithms that interpret sensor streams, generate predictive insights, and issue control commands. A cloud MQTT broker facilitates low-latency, bi-directional publish–subscribe communication and remote control, which helps maintain the synchronization between the greenhouse and the digital twin.

The DT for the greenhouse was made to be photorealistic and behaviorally realistic by rebuilding the virtual world in Unity with C# scripts. Unity's real-time 3D engine allows the sensor data and the control states to be taken from the real-world setup and then put into the interactive simulation, so all the physical parameters get converted into virtual objects that correspond to them. It also supports an intuitive and unobstructed grasp of the system's working, especially in the complicated cases where the number of variables and changes in the environmental conditions are high. Besides that, user immersion is also possible with the help of virtual reality hardware (i.e., VR HMDs and controllers), whereby users can virtually move around the greenhouse, view the different parts of it, and interact with it in a fully immersive context. Such a VR interface serves not only to ease control and visualization but also empowers users to develop scenarios, change parameters, and check the results of their interventions in the virtual world, thus guaranteeing safety if they decide to change the real system later. Multi-tiered graphical user interface with web-based dashboards, mobile apps, and config panels components allows access them. They install a secure communication protocol (WebSocket Secure (WSS)/HTTPS), thus providing real-time system monitoring, continuous feedback, and remote configuration from any location. Besides, a database and analytics layer are introducing the organization's capabilities to collecting historical data, studying the trends and visualizing the performance. This harnessing of analytical power is employed for data-driven decision-making and the enhancement of greenhouse environmental control strategies.

The immersive environment relies on real-time feedback: the DT continuously synchronizes with the Unity engine, while a C# MQTT client ingests live environmental streams: high-frequency updates and physics-based rendering support smooth interaction and in-situ diagnosis without external tools. Built in Unity, the platform runs on VR headsets and conventional devices and connects to a central web server for remote monitoring and control. To mitigate oversight challenges in remote operation, the system provides real-time dashboards for performance metrics, anomaly detection, and scenario-response evaluation; supervisors can also quantify operator efficiency and response times.

Deployment of smart greenhouse solutions

The following services have been implemented in order to illustrate the potential of integrating VR-DT infrastructure for efficiently controlling the IoT devices placed within the smart greenhouse-controlled environment.

Monitoring and managing IoT devices

Through an immersive VR environment, users interact with greenhouse systems in our VR–DTIoT system. IoT-enabled equipment streams real-time data to a central platform, where the VR interface presents an exact digital twin of current conditions. From virtual buttons/sliders, users operate actuators. Each action issues an MQTT command via the broker to the target device; upon execution, both the virtual and physical states update in sync. This closed loop enables rapid, information-driven adjustments. Fig. 3c illustrates actuator control with live state confirmation.

Algorithm 3: User Interaction and System Control in the VR–DT–IoT Greenhouse

```
Data: UserCommands, VRController.Input
Result: Actuator Actions, Updated Digital Twin, User Feedback
/* (0) Rendering & Device Setup                                   */
RenderingEngine ← Configure(UnityRenderer, RealTimeUpdates=True)
Controllers ← Connect({HMD, Controllers})
VisualizationLayers ← Define({Data, Graphs, Alerts, Panels})
/* (1) Main Loop                                                  */
while SystemActive do
    /* Actuator Control                                           */
    if UserCommand = "ActuatorControl" then
        Actuator.Command(Target=Component, Action=UserCommand.Action)
        if ActuatorResponse.Success then
            DTModel.Update(Component=Target, Status=Updated)
            Feedback({''Irrigation Activated'', ''Ventilator Activated''})
        else
            Alert(''Actuator Malfunction'')
        end
    end
    /* VR Input & Navigation                                      */
    VRController.Input ← Read({Joystick, Touchpad, Gestures})
    UserPos.Update(Input, Constraints={Bounds, Collisions})
    VRModel.Navigate(UserPos)
    /* Contextual Interactions                                    */
    if UserInput.Type = "PointAndClick" then
        Overlay.Display(Sensor, {Loc, Value, Hist})
    else if UserInput.Type = "GestureSelect" then
        Chart.Render(Trend(TargetSensor, T=60min))
    else if UserInput.Type = "CommandActuator" then
        Cmd ← Parse(UserInput.Command)
        Actuator.Action(Target, Cmd)
        VRModel.Update(Target, Actuator.Status)
        Feedback.Display(Actuator.Result)
    end
    /* DT{VR Synchronization                                      */
    VRModel.Update(MapTo(SensorData.Live))
    VisualizationLayers.Render(SensorData.Live, {ColorMap, Anim})
    VRModel.Render(DTModel, Layers={Data, Graphs, Indicators})
    /* Subloop & Alerts                                           */
    while UserInteracts do
        UserInput ← VRController.Input()
        if UserInput.Type = "Command" then
            Actuator.Action(Target, UserInput.Command)
            DTModel.Update(Target, Actuator.Status)
        end
    end
    if AlertSystem.Active then
        OperatorAction ← Input(VRController)
        Execute(OperatorAction, Update=VRModel & PhysicalSystem)
    end
end
return Actuator Actions, Updated Digital Twin, User Feedback
```

The VR-DT-IoT integration improves operational efficiency and underpins intelligent greenhouse control. Users also monitor the environment in real time from connected sensors. As shown in Fig. 3d and Fig. 3e, they interact with key components while the VR inset displays the corresponding physical status. Operators can inspect local moisture values, review temperature controls, and, via the DT tool, analyze the last 60 min of trends to support decisions. Integrating live IoT data into VR heightens situational awareness, enables proactive condition management, and ensures seamless coherence between virtual and physical systems.

Interactive visualization and analysis of IoT sensor data in a VR-DT model

The real-time and historical multi-sensor measurements of the greenhouse are the primary sources of data for the integrated analysis of the primary environmental factors in the VR–DT–IoT architecture. The system features a pioneering fusion layer to integrate the IoT data streams, the digital twin model, and the 3D virtual world for the continuous bidirectional synchronization to be possible among physical measurements, virtual images, and human interaction. The data gathered is turned into interactive, context-aware charts in the immersive VR environment. Time series data have been changed to dynamic line graphs and spatial overplots in 3D, where each point is the aggregated or mean sensor readings for a particular time window. Such a visualization method makes it possible to compare temporal and spatial patterns side by side, thus allowing one to graph patterns of, for instance, diurnal temperature cycles and soil moisture fluctuations. People can create these visual properties directly from the virtual model by exploring environmental processes and triggering systemic responses accordingly. This human-in-the-loop operation lowers the level of data visualization to that of experience analytics, thus enabling operators to raise questions, investigate, and respond to sensor data in the virtual model.

Meanwhile, the real-time feedback loop greenhouse operators can have an intuitive and actionable feeling of the system state, thus enabling them to make timely decisions and utilize resources efficiently. Therefore, the visualization system is a graphically represented and novel cyber-physical model of interaction where perception, information, and control merge into one another. The system constantly tracks the activity of a screen sensor over a period of a 60-minute cycle, observing its reaction to outside inputs and ensuing actuation by means of ventilation, misters, pumps, or heaters, as indicated in Fig. 3f. This coming together reflects the platform's ability to bring together monitoring, analysis, and control into one integrated interactive and intelligent space.

Quantitative metrics

As explained in Section 3, the experiment was carried out on a large sample of subjects (N = 23) with no experience in the utilization of VR technology. A total of the subjects was distributed into three groups: an experimental group trained by a virtual reality-based program (N1 = 8), a second one with a computer-based program (N2 = 7), and a control one trained by real equipment (N3 = 8). Subjective data, where opinions and experiences of the participants were recorded throughout the experiment, and objective data, where quantifiable parameters like time taken and errors made during initial, training, and final phases, were the results generated by the experimental process. Single sessions were included in the validation of the proposed framework.

Each session had three stages: briefing, interaction with the VR-DT-IoT model, and evaluation of the virtual experience. During the briefing stage, a presentation of the purpose of the validation sessions, the functional arrangement and the purposes of their interaction with the system was delivered to the participants. Participants were then allowed time to acquaint themselves with and interact with the prototype. It enabled them to investigate the virtual environment, perform functional exploitation, and execute functional evaluation in a simulated greenhouse environment. All groups' performances were compared statistically to determine improvement between the first and the final test in the utilization of the actual greenhouse prototype. The duration taken by each group to finish the control task and the overall number of errors (failures) incurred are provided in the supplementary material (see Supplementary Table S1).

The statistics exhibit significant differences in the error distribution across the groups, indicating a non-homogeneous population. This heterogeneity, brought about by the differences, makes direct comparison of the gains achieved by the experimental group's problematic. Any analysis of the results must, as such, hold these differences constant to obtain an interpretable sense of the initial and final tests. The evaluation activity was concluded by requesting the participants to fill in the SUS questionnaire, for getting quantitative usability feedback regarding the system. The participants were also given a chance to give qualitative feedback by sharing their views and thoughts on their experiences with the VR-DT-IoT model.

After all sessions, data collected were collated and analyzed to validate the framework against primary indicators, including perceived cognitive load, operational efficiency, system usability, and general comprehensibility. Complete statistical results, including mean scores, standard deviations, and score ranges per item, are given in the supporting information (see Supplementary Table S2). This comprehensive analysis completely validated the performance and adherence of the VR-DT-IoT model to user expectations. We suggest calculating the RTI (Eq. (1)) for the first and final assessments so that there is an equal comparison of task completion times between groups. Normalizing the baseline differences in performance enables time distribution to complete the challenge to be more equitably judged.

The bar chart in Fig. 4 presents the mean number of failures made by the groups in each training attempt and offers information regarding error trends over time. The line graph in Fig. 5 illustrates the task efficiency enhancement by showing the average completion times per group over the trials. These metrics enable the effectiveness comparison of the proposed methods in a detailed manner since they clearly reflect accuracy and speed gains along the training process.

$$RTI = \frac{T_{\text{initial}} - T_{\text{final}}}{T_{\text{initial}}} \times 100 \tag{1}$$

We found that the three conditions (Real, VR-DT, and 2D) for controlling the agricultural greenhouse system had different starting times in the first trial. The variations reveal that the training condition influenced participants' baseline performance since participants saw it as a variation in complexity or usability. Employment of the virtual tools (2D and VR-DT) led to more rapid greenhouse control performance, possibly because people did not feel so vulnerable in virtual environments and were therefore more willing to test and

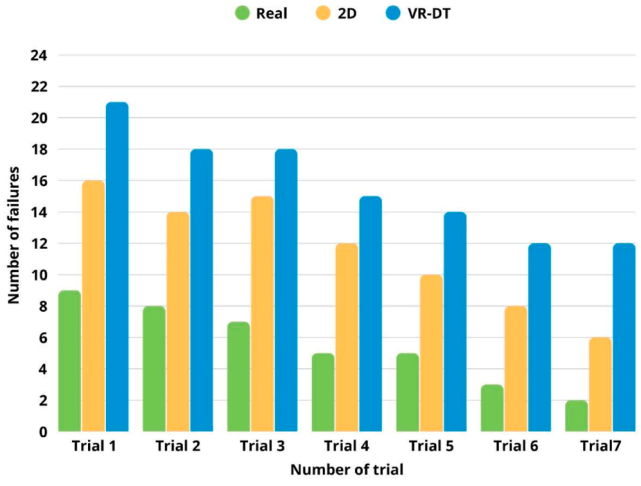


Fig. 4. Average number of failures per group across training trials.

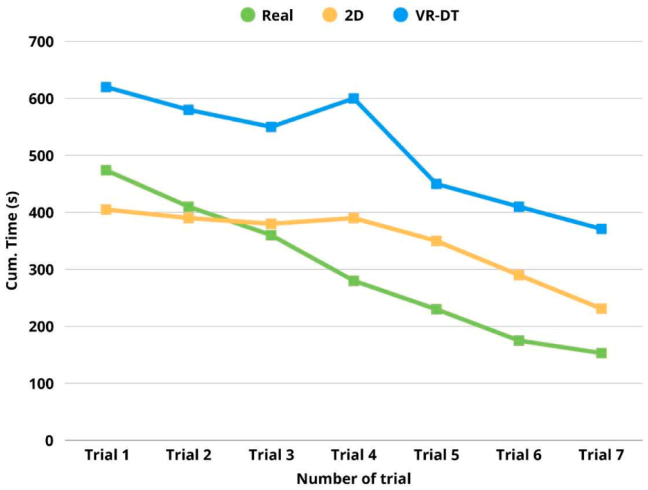


Fig. 5. Average completion times per group across trials.

test system limits. Training inside them became less bounded by physical repercussions of failure involved in real conditions.

The progressive reduction in error rates through repeated attempts in all groups demonstrates the effectiveness of the test procedures in increasing operator capability. However, VR users (VR-DT and 2D) had, on average, higher error rates, presumably because the lower perceived risk enabled them to be more exploratory. Virtual settings can be a trade-off between exploratory learning and task realism.

In the first performances, the 2D group indicated the shortest time (50.6 s), followed by the Real group (67.7 s), and the VR-DT group had the longest time (77.5 s), thus a reflection of the novelty and the mental demands of the virtual environment. With gradual training, all groups could improve noticeably: the Real group shortened its time to 21.8 s, the 2D group to 28.8 s, and the VR-DT group to 46.3 s. Furthermore, the early error frequencies were also highest in the VR-DT group (mean = 2.6), next the 2D group (mean = 2), and finally, the Real group (mean = 1.2). In the final test, all the groups made fewer mistakes than ever before, and the VR-DT group showed the greatest improvement (final mean = 1.5), while the Real and 2D groups scored 0.2 and 0.7 respectively.

To evaluate the effectiveness of each training method, paired *t*-tests were applied to compare groups' initial and final performance in terms of task completion time and number of errors. The paired *t*-test is calculated using the following formula (Eq. (2)):

$$t = \frac{\overline{d}}{S_d/\sqrt{n}} \tag{2}$$

where $\overline{d}$ is the mean of the differences between paired observations, S_d is the standard deviation of these differences, and n is the number of paired samples.

This test statistic determines whether or not the difference between two sets of related measures is considerably larger than zero. By doing so, we carry out a genuine assessment of change within groups across time. Statistically significant improvement in task accomplishment by all three groups was discovered using paired *t*-tests ($p < 0.01$), with the largest improvement occurring among the VR-DT group ($p < 0.001$). As for error, the Real and VR-DT groups both did much better ($p < 0.01$), but the 2D group did so statistically nonsignificantly ($p = 0.072$). What this means is that although optimal performance was achieved by the fastest execution of a task by the Real strategy, best error minimization in the long term was offered by the VR-DT strategy. The less complicated 2D interface provided only marginal benefit in minimizing error. Overall, VR-DT is an efficient training technique in terms of enhanced accuracy for complicated situations with long-term skill and accuracy acquisition at the expense of increased initial learning curve. Table 1 summarizes the time and error generated by the three groups following the three training sessions.

System usability scale (SUS)

The SUS questionnaire aimed to assess the VR-DT environment's existence and utility. The findings in Table 2 provides important

Table 1
Summary of task completion times and failures rates across the three training groups.

Group/ Method	Initial Time *(Mean ± SD)*	Initial Failures *(Mean ± SD)*	Final Time *(Mean ± SD)*	Final Failures *(Mean ± SD)*	%RTI *(Mean ± SD)*
Real	67,71 ± 7,2	1,29 ± 0,8	21,86 ± 2,4	0,29 ± 0,5	67,45 ± 04,4
2D	50,63 ± 8,7	2,00 ± 1,7	28,88 ± 4,8	0,75 ± 0,7	41,33 ± 12,7
VR-DT	77,50 ± 10,7	2,63 ± 0,7	46,38 ± 2,7	1,50 ± 0,5	39,40 ± 06,9

Table 2
Summary of aggregated SUS Scores by participant age and employment status.

Participants	Criteria	Number	Efficiency	Usability
Age	20 - 30	12	3.50	3.94
	31 - 45	7	3.47	3.87
	> 45	4	3.20	3.71
	Total	**23**	**3.39**	**3.84**
Employment	Professor	5	3.62	3.80
	Student	8	3.57	3.85
	Administrator	1	3.22	3.90
	Agricultural field expert	5	3.33	3.95
	Other	4	3.21	3.70
	Total	**23**	**3.39**	**3.84**

information on participant attitudes. With a total average score of 3.84 for usability, the system demonstrates strong user acceptance and a clear understanding of its operational features across most groups. The efficiency score of 3.39, while slightly lower, still indicates that the system aids users in performing tasks effectively, though enhancements could further optimize performance for specific demographics, such as older users or non-technical roles. These findings highlight the system's robust usability and general efficiency, while also pointing to opportunities for improvement in tailoring the VR-DT environment to address the diverse needs of its user base more comprehensively. These results offer a sophisticated perspective on the system's performance in several important areas.

Validation and evaluation

Usability testing incorporated twenty-three participants (students, teachers, and an agricultural field expert), who were introduced to the system using real-world scenario simulations. The participants had to explore the environment in simulation mode, interpreting the sensor readings. Such tests yielded positive feedback to the commissioning team as they focused on user-friendliness and simplicity, which were test specimens. Moreover, the test is a good future suggestion for integrating all the systems in greenhouse activities.

A comparative study also helped in evaluating the performance of the system compared to traditional greenhouse management protocols. Results declared that the integrated VR–DT–IoT platform brings a new dimension to operational intelligence due to the fact of mere combination of awareness of data in real-time and ease of manipulation by humans; hence, virtual visualization and physical interaction are united.

The research results showed that through the system, decision-making was enhanced as the users were provided with real-time information; in addition, user satisfaction was reportedly enhanced through ease of the VR interface. The comparison brought out the system's advantages vis-à-vis the conventional means, with the prototype alluding to the possibility of process reduction and decision-making simplification. The participants' response empowered the system. Through an iterative process, such elements as the inclusion of additional gesture controls in the VR navigation element and the provision of more nuanced controls to display sensor data were added to the system to integrate user recommendations. These additions complemented the performance and efficiency of the system further to the needs of users, a characteristic of its functionality in reality to use a greenhouse.

System Usability Scale (SUS) was utilized to gather self-report system usability ratings of subjects on an extensively reported survey, which yielded quantifiable user satisfaction scores.

Parallel measures of performance on, say, RTI were calculated in order to yield quantitative indices of learning gain in task performance across sessions. Lastly, overall number of failures and time on task, group means, provided objective indices of system usability and learnability. Through this testing and validation, it was carried out in a way that the VR-DT-IoT system not only manages to perform as required but also supports pragmatic usability needs, and it was established to be viable for real-world application in agricultural greenhouse management.

Discussion

Contextual relevance and research gap

This study demonstrates that coupling a Virtual Reality–enabled Digital Twin with IoT in a bidirectional, real-time loop changes how greenhouse operations are learned, executed, and audited. Prior efforts in digital agriculture ontologized either the DT side (analytics, prediction) or the interaction side in isolation.

Recent DT-greenhouse studies have validated climate monitoring and predictive models in South Africa and Europe, but largely as unidirectional data-to-model pipelines without immersive, human-in-the-loop control [36,37]; robotic harvesting work has leveraged DT for planning yet remained task-specific and not operator-centric [38]. Conversely, VR research in greenhouses has focused on comfort, interaction design, and educational affordances, again without direct coupling to physical actuators or live sensor streams [23,39].

Our results close this gap by evidencing a coherent, latency-aware cyber-physical stack where operators interpret spatialized data and command actuators in the same workflow—an integration also aligned with broader Industry 4.0 sustainability vectors highlighted for built environments [40].

Interpretation of key findings

Failures decreased monotonically across seven trials in all modalities (Fig. 4), with VR-DT showing the steepest early decline, which aligns with immersive environments facilitating proceduralization and spatial reasoning. Time also went down cumulatively over the trials (Fig. 5), with Real finally scoring the lowest end time (Table 1). At the same time, VR-DT made significant error reductions (from 2.63 ± 0.7 to 1.50 ± 0.5) and considerable time improvements (RTI $39.4 \pm 6.9\%$). This sequence of events reveals the complementary strengths that physical labs can speed up under routinized familiarity, while immersive VR-DT concentrates on situation awareness and decision accuracy during the learning phase—exactly when risk and uncertainty are at their highest.

The SUS composite is high (3.84/5 overall; Table 2) and consistent across different age groups and employment statuses, thus reflecting strong acceptability and perceived effectiveness. Slightly lower efficiency scores for older users (3.20 for >45) provide actionable UI improvement indications. The findings are an extension of VR-DT usability questionnaires that link comfort/playability to user ratings [23] by incorporating task-level evidence (time, errors) that bridge human-factors measures and operational performance.

Comparison with recent literature

Greenhouse DT monitoring and prediction (e.g., SVR-based temperature regulation under climate uncertainty in South Africa; IoT living-lab pipelines in Ireland) justified feasibility and established the foundations for predictive analysis [36,37], while DT-empowered robotic harvesting optimized task performance via virtual planning [38]. Our platform extends these routes further by:

1. providing interactive, two-way synchronization with physical actuators in real time;
2. embedding immersive visualization elevating operator cognition (presence, awareness) from voluntary add-on to integral control surface;
3. providing human-subject data ($N = 23$, seven experiments) that links the DT stack to error reduction and timing benefits—thus moving from "DT as a model" to "DT as an operational medium."

On the interaction side, VR greenhouse work has focused on comfort and interface variants [23,39]. We substantiate those issues but show that, under the right pacing and UI, immersion does go with performance gain, giving the missing behavioral bridge between operations and interface design. Finally, Industry 4.0 technology convergence reviews argue that they attain sustainability value when DT, IoT, and immersive tech are integrated end-to-end [40]; our empirical data actualize that contention in an agricultural and Africa-specific context.

Significance for Africa and policy relevance

For African controlled-environment agriculture—often constrained by water and energy scarcity, skills gaps, and variable climates—a deployable VR–DT–IoT stack offers three immediate utilities:

(i) Precision environmental control via spatial alerts and closed-loop actuation (supports SDG 2.4: sustainable production through resilient practices);
(ii) Digital infrastructure for decisions—modular MQTT/Node-RED pipelines and topic semantics (supports SDG 9.1: reliable, sustainable infrastructure for data-driven operations);
(iii) Climate-risk readiness—faster anomaly detection, safer rehearsal of fault scenarios, and reduced waste (supports SDG 13.1: strengthened adaptive capacity).

These attributes align with Agenda 2063 priorities for modern high-productivity agriculture, and provide a replicable template for national smart-farming programs (training curriculae, extension services, and technology transfer) minus the capital intensity of fully automated farms.

Conclusion

This research has shown how these three technologies can be combined to significantly improve the monitoring, control, and decision-making processes of agricultural greenhouses. Moving beyond the current limitations of solutions that only allow for either data acquisition or offline simulation, this paper presents a new cyber-physical platform that makes it possible to interact in real-time, to get an immersive experience, and to have two-way control with the physical and virtual worlds.

The innovative VR-DT-IoT system achieves this by delivering real-time data from IoT-powered actuators and sensors to interactive visualization and interaction in a virtual 3D space. The integration enables greenhouse managers to monitor critical environmental parameters visually and, through the interactive system, to have a dialogue with the components for ventilation, irrigation, and lighting control at a micro-level. The system, via well-planned usability tests with a panel of experts from different functional backgrounds, has worked out in operational work flows, environmental control, and data-driven decision-making, which were simpler in form and understanding than before. Volunteers spoke exceptionally highly of how quickly and clearly the virtual interface responded to them and how easily the system could physically control the actuators through their digital counterparts. The system does create a

new technology norm by integrating these features—predictive modeling, real-time synchronizing, immersive control—into a single shared human-in-the-loop design, an unambiguous advance in farm digitization.

This is a crucial and basic step in Africa to move towards development of robust, technology-based agriculture systems which can confidently meet the "old problems" challenge. The VR-DT-IoT system is an inexpensive, simple tool whose introduction is consistent with the aims of precision agriculture and resource efficiency. Doing so clears the way and makes the transition to Agriculture 5.0 possible, which is characterized by more innovative, interactive, and productive greenhouse systems. Besides, further research is needed to support artificial intelligence for predictive analysis and decision-making and to make the platform more common across different agriculture domains with more field validation and user testing.

CRediT author contribution statement

Hicham Slimani: Methodology – Conceptualization – Data curation – Writing – original draft – Software – Hardware – Resources – Funding acquisition – Validation. **Abdelilah Jilbab**: Methodology – Project administration – Supervision – Resources –Writing – original draft – Software – Formal analysis – Validation. **Jamal El Mhamdi**: Supervision – Methodology – Writing – review editing – Resources – Software – Visualization – Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.sciaf.2026.e03665.

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